



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University

Accredited by NAAC & An ISO 9001:2015 Certified Institution

NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Computer Science and Business systems

Academic Year 2023-2024 (Even Semester)

Degree, Semester & Branch: IV Semester B.Tech -CSBS

Course Code & Title: AL3451-Machine Learning

Name of the Faculty member (s): Mrs. B.Yazhini

Innovative Practice Description

Unit / Topic: UNIT II /Supervised Learning/Logistic Regression

Course Outcome: CO2

Learning Outcome: 3,4

Activity Chosen: Think Pair Share

- **Justification:**

- ✓ Think pair share activity involves the students to think individually and to share the knowledge with classmates.
- ✓ In python, control flow helps to help the order of an execution of program based on the values and logic.
- ✓ The student's individual solving capability can be tested and the gained ideas can be shared by think-pair-share activity

- **Time Allotted for the Activity:** 30 Minutes

- **Details of the Implementation:**

A Collaborative, active learning strategy, in which students work on a question posed by instructor, first individually (Think), Write in a paper, then work in pairs or groups (Pair) to solve the problem, and finally share their solution with the entire class (Share).

- **Think** – Write- Students were provided with questions on generic methods. The student has to think individually and write the answer in a paper.

- **Pair**- After writing the answers, the students were allowed to discuss with neighbor and justify the resultant answer to each other. The pair with neighbor was shown in Fig.2

- **Share** - Once the conclusion is made between the pair, then it is to be shared among the whole class by any of the student from the team and it is shown in the Fig 2.

CO-PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO9	PO11	PO12
CO2	3	3	3	2	2	2	2	2

CO	PSO1	PSO3
CO2	3	2

Innovative practice	POI 3	PO2 3	PO3 3	PO4 2	PO5 2	PO9 2
Justification for correlation	Students will be able to apply supervised learning algorithms for making predictions	Students can be able to identify and analyze the problem	Students can be able to design supervised machine learning algorithms for solving the problem	Students will apply research-based knowledge and methodologies to validate the effectiveness and performance of various Machine learning algorithms	Students will deploy the linear regression, classification and generative models.	Students will work effectively as individuals and as members of teams to develop and deploy machine learning models in real world scenario.

PO mapped:

Innovative practice	PO11 2	PO12 2
Justification for correlation	Students will manage projects related to Decision Tree Regression, Naïve Bayes and Support Vector machine	Students will be encouraged to explore emerging Machine learning application through continuous learning and research

PSO Mapped:

Innovative practice	PSO1 2	PSO3 2
Justification for correlation	Students will manage projects related to Decision Tree Regression, Naïve Bayes and Support Vector machine	Students will be encouraged to explore emerging Machine learning application through continuous learning and research

- Images / Screenshot of the practice:

Logistic regression. Think, Pair, Share

Shaly (42)
Gayathri
Meena (12)

Logistic regression is a statistical model and it is used to data analysis technique that uses mathematics to find the relationship between two factors.

In linear regression, the values lies other than 0 & 1 also. So logistic regression lies between 0 & 1.

In logistic Regression, there exists Sigmoid function.

The graph can be represented as

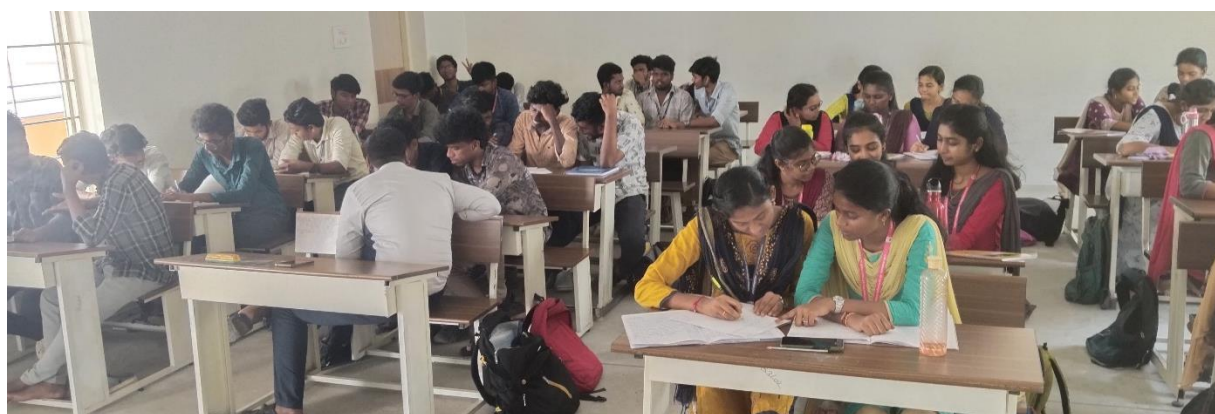
Sigmoid function = $\frac{1}{1+e^{-z}}$, $z = mx+c$

Application:

- In medical field
- Stock market and grading System
- Airline
- Social Media & network

There are two factors x and y. Using logistic regression find the relationship between x and y.

Figure 1 & 2 THINK PAIR SHARE Sample Screen shots



Reflective Critique:

❖ Feedback of practice from students and other stakeholders:

- Some of the students write answers for the questions easily and share their ideas with others.
- Some of them need more explanation and examples to get clarity on the generic methods.

❖ Benefit of the practice: (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

It helps the students to share ideas with classmates and builds oral communication skills. They completed the task by coordinating with each other and get more confidence, clear understanding in this topic when discussing with their peers.

❖ **Challenges faced in implementation:**

- The students who didn't understand the concept clearly is not able to complete it within time.
- Making all the students to involve and present actively in this activity is a challenging.

References:

- ❖ https://www.ritrjpm.ac.in/images/computer-science/14_CS8392_TPS.pdf
- ❖ https://www.ritrjpm.ac.in/images/computer-science/2022-2023/3_GM_CS3353_Evaluation_Expression_TPS.pdf
- ❖ https://www.ritrjpm.ac.in/images/computer-science/CS8451-Think-Pair-Share_PJT.pdf

Faculty-In-Charge

HOD/CSBS